

Log-Location-Scale Survival Models for Heart Failure

Data: A Comparative Study

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Introduction

Heart failure is a major cause of mortality worldwide, with highly variable survival patterns. Modeling time-to-event data in such settings requires flexible approaches to capture survival behavior and underlying risk factors.

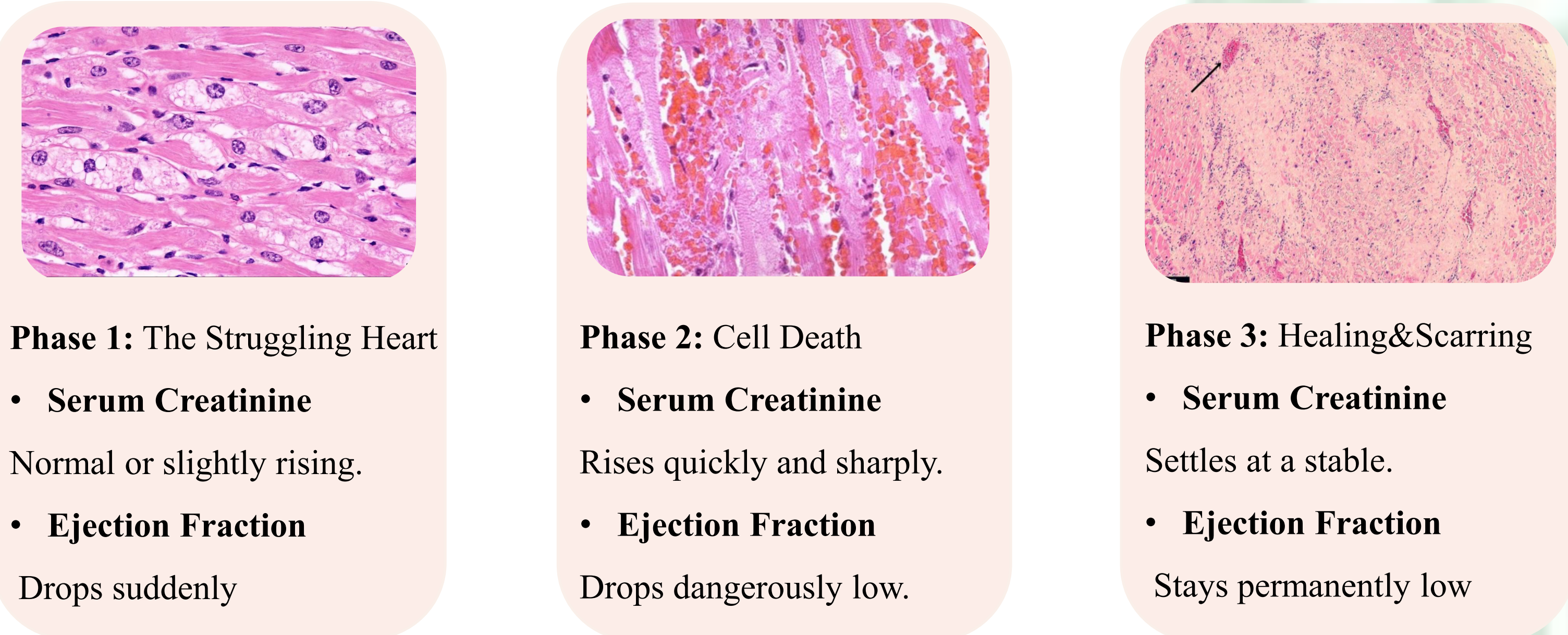


Figure1: Three-phase histopathological progression of myocardial damage from cell stress (Phase 1) and necrosis (Phase 2) to tissue repair (Phase 3) (Ghafoor et al., 2020).

This study applies parametric survival models within the log-location-scale framework—specifically the Weibull [2], Log-normal [3], and Log-logistic [4] distributions—to identify key predictors and compare model performance.

Methodology

Dataset: Heart Failure Clinical Records

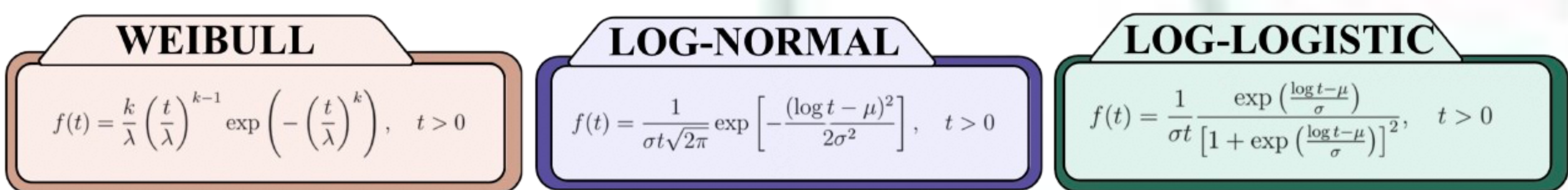


Model Framework:

Survival time is modelled using the Accelerated Failure Time (AFT) approach within the log-location-scale (LLS) framework:

$$\log(T_i) = \mathbf{x}_i^T \boldsymbol{\beta} + \sigma \varepsilon_i, i = 1, 2, \dots, n,$$

where $\boldsymbol{\beta}$ represents regression coefficients, σ is the scale parameter, and ε_i follows a specific distribution.



Model Specification and Evaluation:

- Variable selection: **Backward selection.**
- Model comparison: **AIC, BIC, CAIC** and **Log Likelihood.**
- Model adequacy: **Kaplan–Meier Survival Estimate** and **Cox–Snell residuals.**

Results and discussion

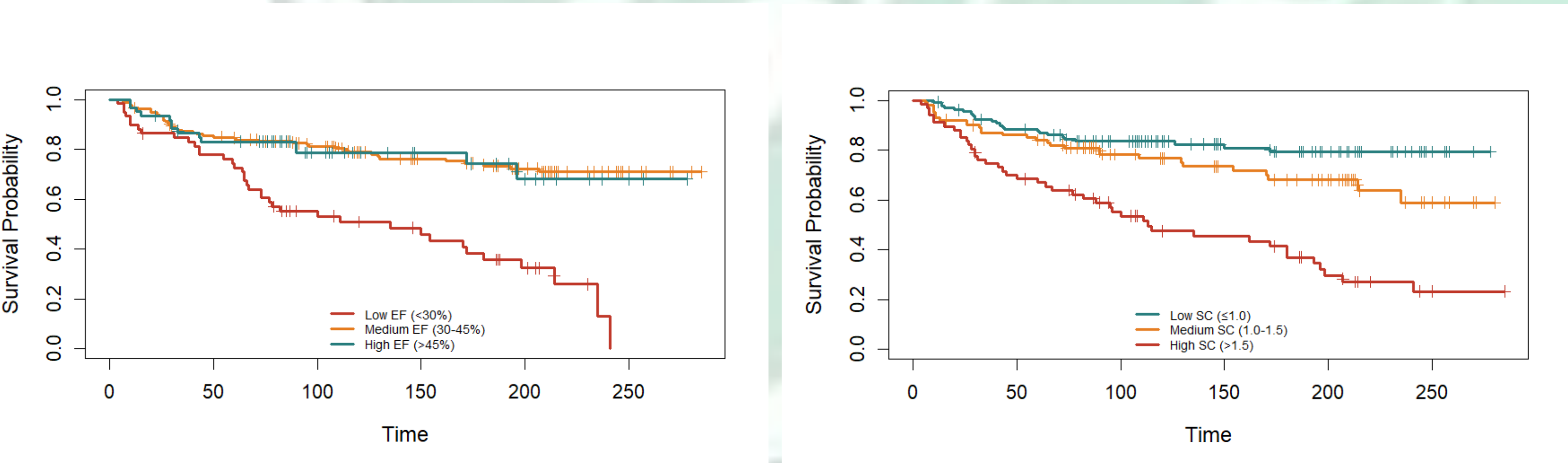


Figure2: Kaplan-Meier by Ejection Fraction Figure 3: Kaplan-Meier by Serum Creatinine

- Kaplan–Meier curves show that survival is lower for patients with low Ejection Fraction and high Serum Creatinine, with both groups exhibiting a noticeably sharper decline compared to the others.

Model results and Validation:

Table1: Parametric Survival Model Comparison Statistics

MODEL	LOGLIK	AIC	BIC	CAIC
WEIBULL	-630.812	1277.624	1307.227	1315.227
LOG-NORMAL	-633.279	1282.557	1312.161	1320.161
LOG-LOGISTIC	-632.179	1280.359	1309.962	1317.962

- The Weibull model has the lowest AIC, BIC, Log Likelihood, and CAIC, confirming it as the best-fitting model.

Table2: Weibull Regression Model Coefficients

Variable	coefficient	Std.Error	p-value
Age	-0.04743	0.00971	0.001
Ejection Fraction	0.05043	0.01132	0.001
Serum Creatinine	-0.31606	0.07277	0.001

- The Weibull regression identifies Age, Ejection Fraction, and Serum Creatinine as the most significant predictors.

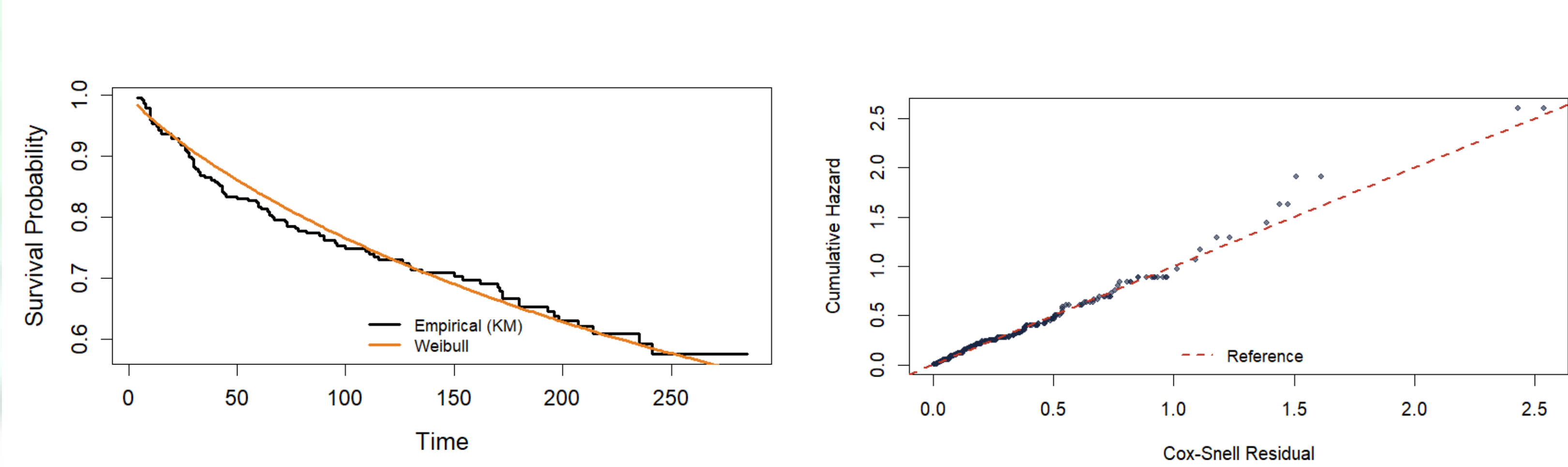


Figure 4: Kaplan-Meier Weibull fitted & empirical Figure 5: Cox-Snell — Weibull

- The Weibull survival curve closely matches the KM estimate, and the Cox–Snell residuals follow the reference line, indicating an excellent overall model fit.

Conclusion

The Weibull model provides the best overall fit for the heart failure data, outperforming the Log-normal and Log-logistic models. Age and Serum Creatinine are associated with reduced survival time, while Ejection Fraction has a protective effect, highlighting the importance of both cardiac and renal function in patient prognosis.

Acknowledgements

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References

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